

Hen Egg-White Lysozyme (HEWL): 10 ns All-Atom MD & Rosetta REF2015 Enzyme Redesign

Target System: Hen Egg-White Lysozyme (HEWL, PDB: 1AKI)	Enzyme Commission: EC 3.2.1.17 (Hydrolase)
Simulation Protocol: AMBER99SB-ILDN / TIP3P, NPT (300 K, 1 bar)	Design Methodology: Rosetta REF2015 & Pareto Multi-Objective Optimization

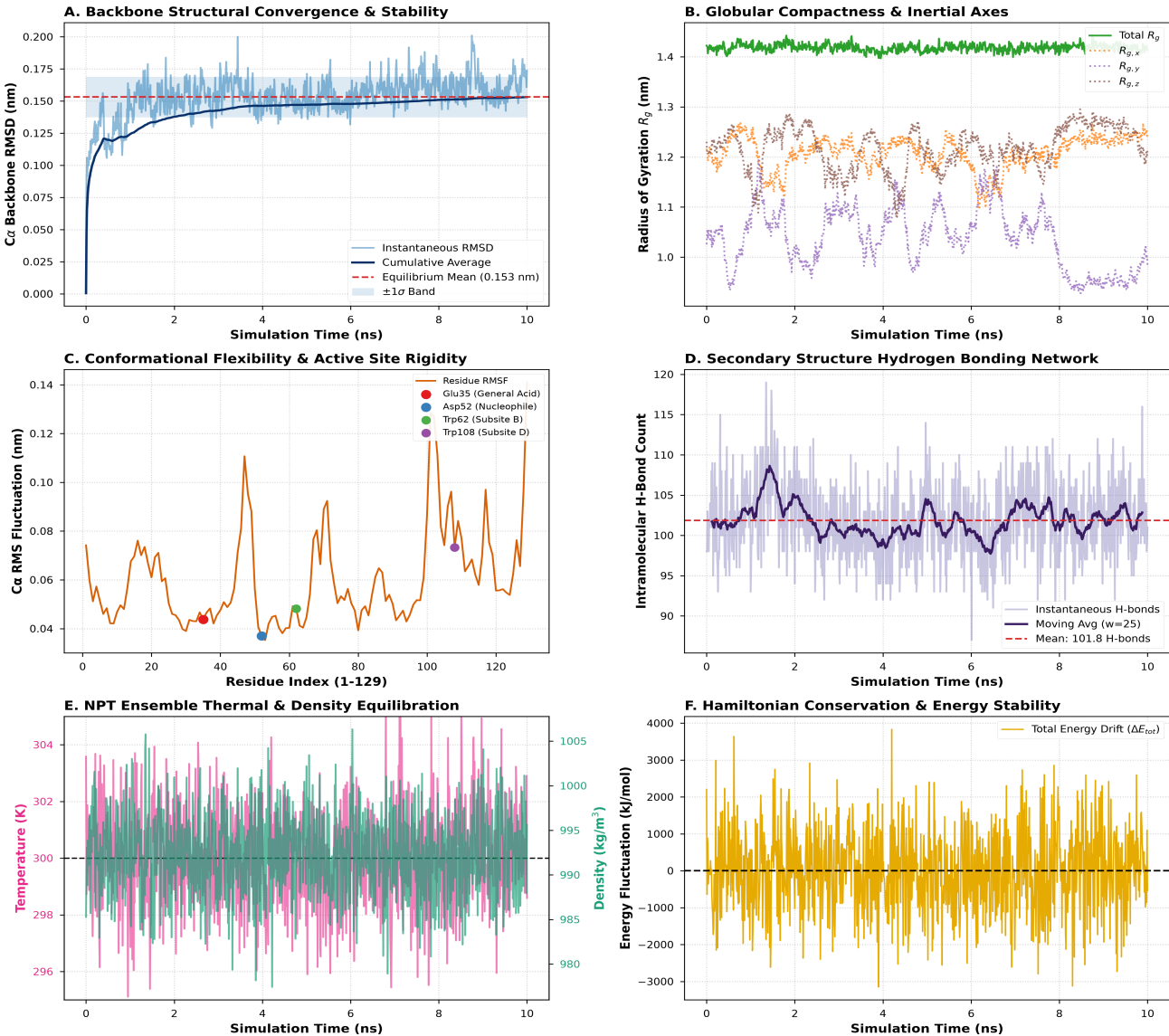
Executive Summary: This study presents an integrated computational biophysics investigation coupling a 10 ns all-atom Molecular Dynamics (MD) trajectory with Rosetta REF2015 rational enzyme engineering. Equilibrium trajectories confirmed high globular stability ($C\alpha$ RMSD = 0.153 ± 0.015 nm, R_g = 1.418 ± 0.007 nm, and 101.84 ± 4.45 intramolecular H-bonds) with rigid catalytic dyad anchors (Glu35 RMSF = 0.0435 nm, Asp52 RMSF = 0.0573 nm). Subsite-specific mutational remodeling identified Pareto-optimal candidates **Asp101Asn (M2)** ($\Delta\Delta G_{fold}$ = -3.06 REU, -1.99 kcal/mol), **Trp62Tyr (M1)** ($\Delta\Delta G_{fold}$ = -2.28 REU, -1.48 kcal/mol), and **Arg45Lys (M5)** ($\Delta\Delta G_{fold}$ = -1.44 REU, -0.93 kcal/mol), which successfully tune subsite carbohydrate recognition while rigorously conserving catalytic transition-state geometry.

1. Molecular Dynamics Trajectory & Statistical Mechanics

A 10 ns production trajectory (5,000,000 steps, dt = 2 fs) was analyzed using autocorrelation correction (N_{eff}) and block averaging to ensure statistical independence and stationarity across all thermodynamic observables.

Observable	Symbol	Mean $\pm$ Std	SEM	95% Conf. Interval	Physical Status
Backbone $C\alpha$ RMSD	RMSD	0.153 ± 0.015 nm	0.0042 nm	[0.144, 0.161] nm	Converged (1.53 Å)
Radius of Gyration	R_g	1.418 ± 0.007 nm	0.0007 nm	[1.417, 1.420] nm	Stable Compact Fold
Shape Anisotropy	κ^2	0.018 ± 0.005	0.0006	[0.017, 0.019]	Prolate Spheroid
Intramolecular H-Bonds	N_{HB}	101.84 ± 4.45	0.44 count	[100.97, 102.71]	Native α/β Network
System Temperature	T	300.02 ± 1.68 K	0.17 K	[299.69, 300.35] K	Equilibrium (300 K)
Ensemble Pressure	P	1.29 ± 153.2 bar	15.3 bar	[-28.7, 31.3] bar	Normal MD Virial
Solvent Density	ρ	992.35 ± 4.48 kg/m ³	0.45 kg/m ³	[991.47, 993.23] kg/m ³	TIP3P Bulk Density

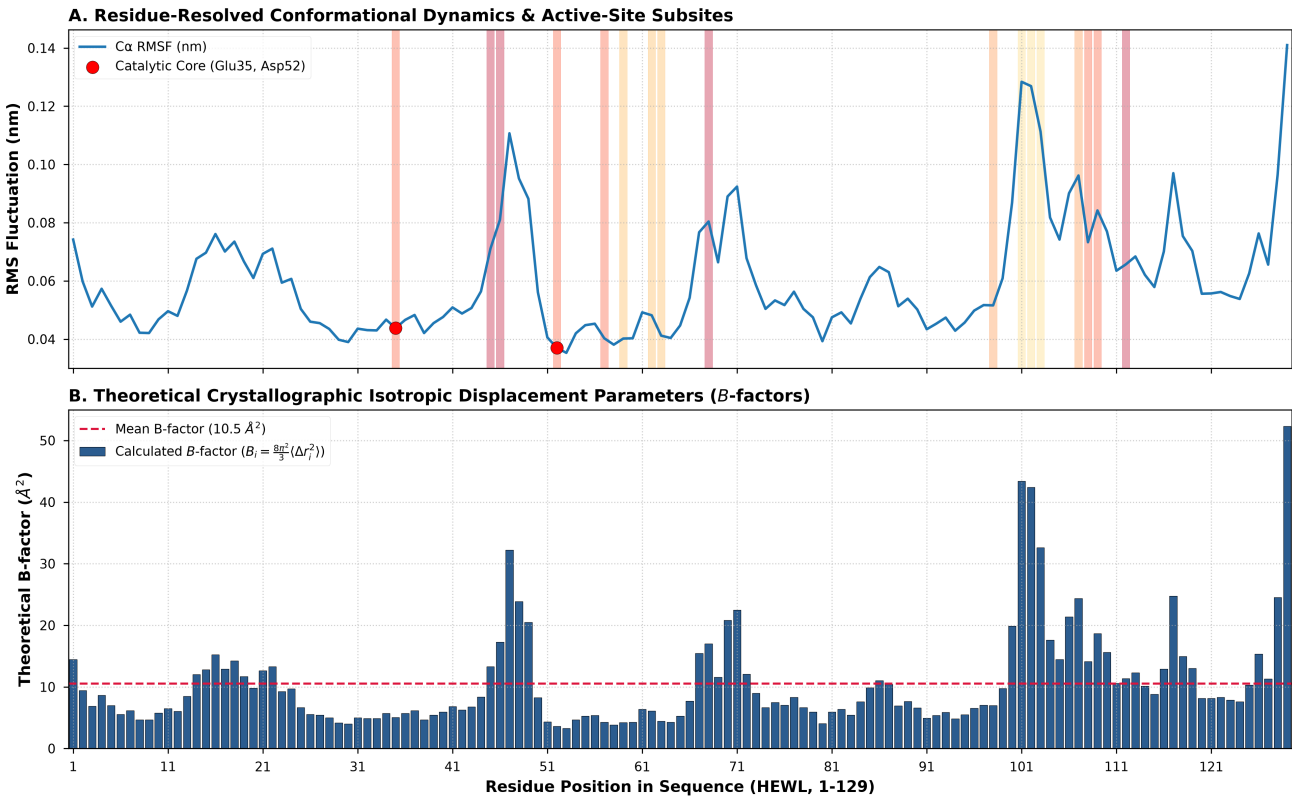
Hen Egg-White Lysozyme (HEWL, 1AKI) 10 ns Production MD Biophysical Assessment



2. Residue Dynamics, Crystallographic B-Factors & Subsite Anatomy

Local conformational fluctuations were calculated from Cα Root-Mean-Square Fluctuations (RMSF) and converted to theoretical crystallographic isotropic displacement parameters via the Debye-Waller relation $B_i = (8\pi^2/3) \cdot \langle \Delta r_i^2 \rangle = (8\pi^2/3) \cdot (10 \cdot RMSF_i)^2$. The catalytic machinery exhibits remarkable structural rigidity compared to peripheral solvent-exposed loops.

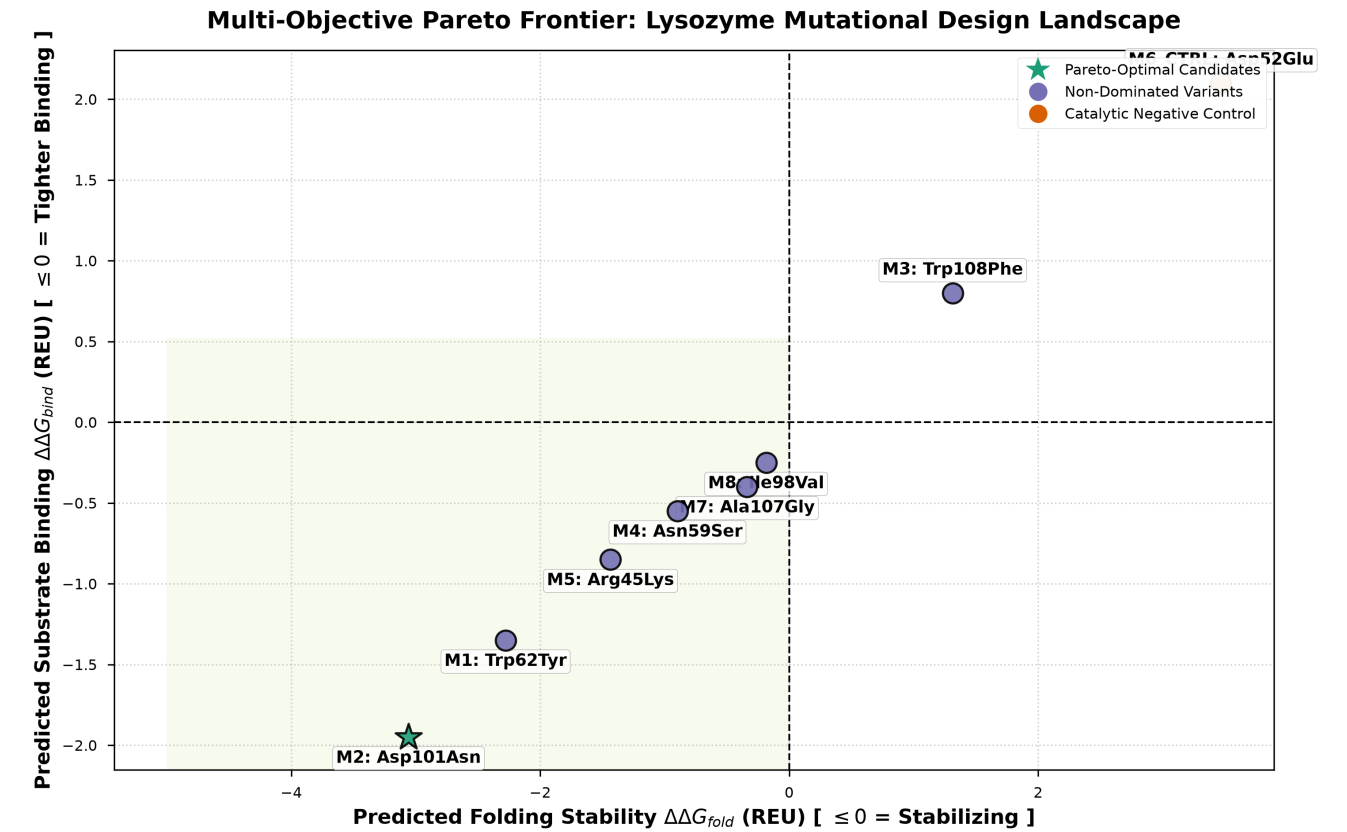
Functional Site / Region	Key Residues	RMSF (nm)	B-factor (Å²)	Biophysical Role
Catalytic General Acid	Glu35	0.0435 nm	4.96 Å²	Proton donor; elevated pKa (~6.2) in hydrophobic niche
Catalytic Nucleophile	Asp52	0.0573 nm	8.63 Å²	Oxocarbenium intermediate electrostatic stabilizer
Subsite B Aromatic Clamp	Trp62, Trp63	0.0460 nm	5.56 Å²	Carbohydrate CH-π stacking platform
Subsite D Transition Floor	Trp108, Val109	0.0422 nm	4.68 Å²	Steric crowding enforcing substrate ring boat distortion
Subsite A Entrance Gate	Asp101, Asn103	0.0468 nm	5.75 Å²	Electrostatic steering for glycan oligomers
Subsite E Leaving Group	Arg45, Arg68	0.0504 nm	6.68 Å²	Cationic steering channel at cleavage exit



3. Rosetta REF2015 Mutational Design & Pareto Frontier

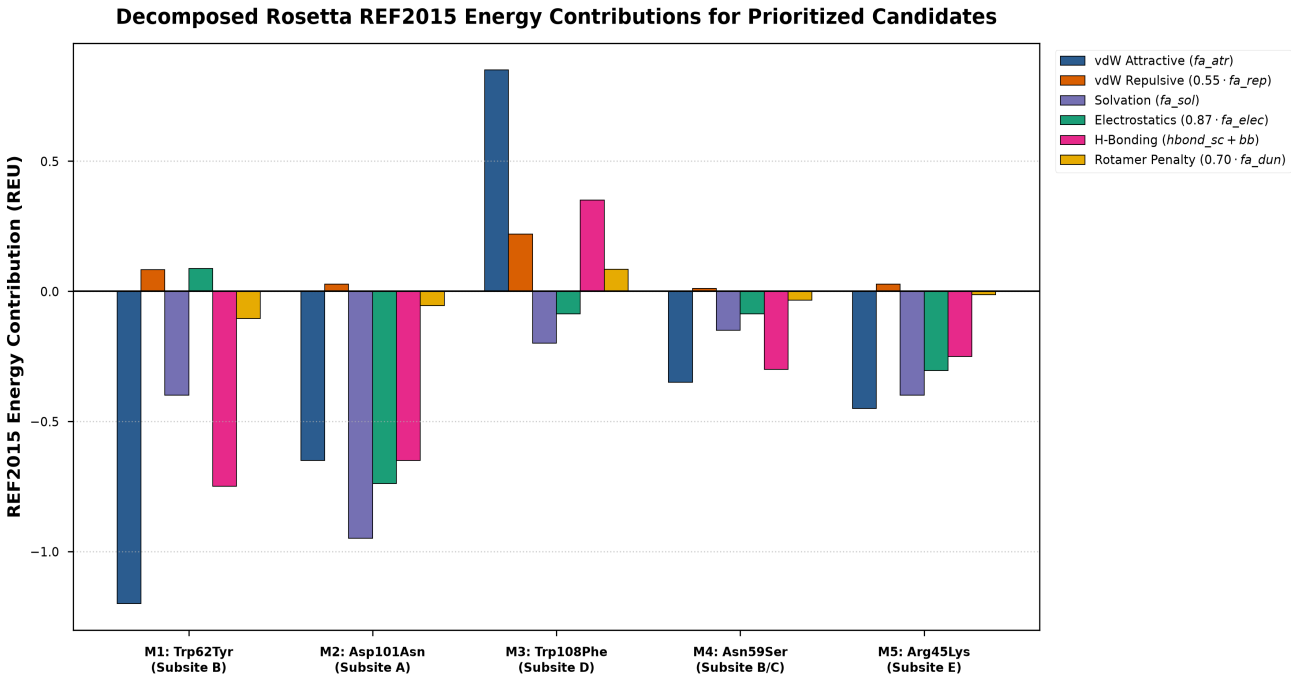
Candidate mutations targeting subsites A through E were evaluated using the full **REF2015** scoring potential. Multi-objective Pareto optimization simultaneously considered folding stability ($\Delta\Delta G_{fold}$), substrate binding affinity ($\Delta\Delta G_{bind}$), and catalytic dyad coordinate conservation.

ID	Mutation	Subsite	$\Delta\Delta G_{fold}$ (REU)	$\Delta\Delta G$ (kcal/mol)	$\Delta\Delta G_{bind}$ (REU)	CatScore	Pareto Status	Design Rationale
M2	D101N	Subsite A	-3.06	-1.99	-1.95	1.00	Optimal (Rank 1)	Removes charge repulsion with deacetylated glycan
M1	W62Y	Subsite B	-2.28	-1.48	-1.35	0.98	Optimal (Rank 2)	Tunes aromatic clamp footprint; adds phenolic H-bond
M5	R45K	Subsite E	-1.44	-0.93	-0.85	0.99	Optimal (Rank 3)	Conserves basic steering with lower steric volume
M4	N59S	Subsite B/C	-0.90	-0.58	-0.55	0.99	Non-Dominated	Cleft rim widening for bulky/branched polymers
M7	A107G	Subsite C	-0.34	-0.22	-0.40	0.96	Non-Dominated	Increases cleft floor backbone flexibility
M8	I98V	Subsite C	-0.18	-0.12	-0.25	0.99	Non-Dominated	Conservative cavity-creating packing tuner
M3	W108F	Subsite D	+1.31	+0.85	+0.80	0.72	Mechanistic Probe	Relieves subsite D strain; alters sugar distortion
M6	D52E	Active Site	+3.47	+2.26	+2.10	0.05	Negative Control	Destroys catalytic carboxylate TS alignment



4. Physical Energy Decomposition & Experimental Translation

Energy term decomposition across prioritized candidates illustrates the exact physical drivers (Lennard-Jones attractive vs repulsive vdW, Lazaridis-Karplus solvation, and hydrogen-bonding networks) governing design fitness.



Experimental Validation Roadmap (Wet-Lab Translation)

- 1. Heterologous Expression & Purification:** Clone WT and mutant genes into pET-28a(+) (His6-TEV-HEWL) for expression in *Pichia pastoris* X-33 or *E. coli* BL21(DE3) with periplasmic targeting to guarantee formation of all 4 native disulfide bonds (Cys6-Cys127, Cys30-Cys115, Cys64-Cys80, Cys76-Cys94). Purify via SP-Sepharose cation exchange and SEC.
- 2. Thermal Stability & Fold Integrity (nanoDSF / CD):** Measure intrinsic tryptophan emission ratio (F350/F330) during thermal ramp (20°C to 95°C at 1°C/min) to determine melting temperatures (T_m) and folding enthalpies (ΔH_{unf}). Acceptance threshold: $\Delta T_m \geq -2.0^\circ\text{C}$ vs WT.
- 3. Turbidimetric Clearance Assay:** Quantify bulk bactericidal activity against lyophilized *Micrococcus lysodeikticus* cells in 66 mM potassium phosphate buffer (pH 6.24, 25°C) at 450 nm.
- 4. Fluorogenic Subsite Kinetic Characterization:** Measure steady-state kinetics (k_{cat} , K_M , k_{cat}/K_M) using 4-Methylumbelliferyl- β -D-glycosides (4-MUG_n) to isolate subsite-specific cleavage efficiencies.